

SIMT-Aware Lockstep Verification and Functional-Coverage Closure Methodology for an Open-Source RISC-V GPGPU: A Complete UVM 1.2 Environment

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Abstract

Open-source RISC-V GPGPUs ship without reference-model verification, functional-coverage models, or sign-off discipline, while concurrent fuzzing work demonstrates that real defects populate exactly this design class. This paper presents a complete UVM 1.2 verification environment and closure methodology for the Vortex RISC-V GPGPU, and the findings it produced on both sides of the comparison. The environment wraps a bus-master SIMT device with role-inverted agents, integrates the project's functional simulator as a per-configuration golden model over DPI-C, and renders verdicts through two checkers that are themselves qualified by permanent fault-injection guards: a bidirectional end-state scoreboard and a per-instruction, per-lane lockstep comparator governed by five alignment rules that we semi-

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formalize as precondition–soundness properties. A two-pass load-value feed makes fenceless multi-core programs instruction-granularity verifiable—verified modulo the fed race resolutions, with the assumption stated and, in the flagship case, discharged by a trace-level race witness. A three-layer coverage model adds what is, to our knowledge, the first published SIMT functional-coverage layer for RTL GPU verification (IPDOM divergence depth, scratch-pad bank conflicts, coalescing classes), closed under machine-generated, RTL-cited exclusions enforced by a blocking waiver-integrity gate. The methodology surfaced externally corroborated RTL defects (a JALR LSB ISA deviation, a reset-relay X window behind a silenced assertion) and reference-model defects found by the lockstep itself. The checking stack is cheap to operate: lockstep adds no measurable wall-clock overhead at either end of the kernel-size spectrum and roughly 30% peak memory at scale. Stimulus *kind*, not volume, moves the coverage model: ninety additional scalar-generator programs closed zero bins, while fifty-seven axis-directed programs closed the SIMT targets.

Keywords: UVM, RISC-V, GPGPU, SIMT, functional verification, lockstep co-simulation, functional coverage, open-source hardware

1. Introduction

Open-source silicon has an industrial verification problem with a sharpening edge. The RISC-V ecosystem maintains mature verification assets for scalar cores—step-and-compare environments, constrained-random generators, ISA coverage libraries [5, 7, 6, 9, 4, 10]—but its most complex open processor class, the GPGPU, has had none of this. Vortex [1, 2], the leading

open RISC-V GPGPU, ships a directed-kernel regression with no RTL-level reference checking and no functional-coverage model. Concurrently, FuzzGPU [3] demonstrated that a fuzzer can find real defects in exactly this class of design (20 previously unknown bugs, 10 CVEs, across Vortex and Ventus), converting the gap from an academic inconvenience into a security exposure.

This paper closes the other half of that problem: not bug discovery, but *coverage-quantified sign-off*. Its contributions:

1. a complete, configuration-parametric UVM 1.2 [11] environment for a bus-master SIMT GPGPU (Section 3);
2. per-instruction, per-lane lockstep for SIMT via five alignment rules, *semi-formalized* as precondition–soundness properties (Section 5);
3. a two-pass load-value feed that makes fenceless multi-core programs instruction-granularity verifiable, with its soundness assumption stated and, where possible, discharged by trace-level race witnesses (Section 6);
4. a three-layer coverage model including the first published SIMT functional-coverage layer for RTL GPU verification (Section 8);
5. an exclusion methodology—machine-generated, RTL-cited waivers under a blocking hits-invariant merge gate—elevated here to a named, reusable contribution (Section 9);
6. an evidence-classified findings register covering both the RTL and the reference model, with dispositions and external corroboration (Sections 10–10.2), and the verification-cost and per-generator marginal-coverage evidence that adoption requires (Section 11).

All honesty constraints of the conference version carry over verbatim (Section 12); every number in this paper traces to a banked artifact or is marked as pending.

2. Background and related work

2.1. The DUT and its reference models

Vortex organizes as clusters $\rightarrow$ sockets $\rightarrow$ cores $\rightarrow$ warps $\rightarrow$ threads with per-core caches, optional shared L2/L3, an immediate-post-dominator (IPDOM) reconvergence stack, and hardware barriers; the verified build is RV32IMAF at a pinned revision on QuestaSim. Vortex has no trap architecture—a property with verification consequences (Section 10). SimX, its functional simulator, is structurally an RVVI-shaped golden model [6]; Spike [8] cannot execute the SIMT instructions and serves only as a scalar, independent base-ISA cross-check (11,076-retirement three-way audit: zero mismatches, injection-proven non-vacuity). *The SIMT axis has no independent reference*—a structural ceiling we return to in the threats section.

2.2. Reference-model verification of processors

Step-and-compare is the standard of care for RISC-V CPUs. core-verif [5] established the open-source pattern with ImperasDV [9] over RVVI [6]; riscv-dv [7] supplies constrained-random programs with a trace-compare flow; DiffTest [4] industrialized per-instruction architectural comparison over DPI-C for XiangShan; OpenTitan [10] demonstrates sign-off-grade UVM on open silicon at SoC scale. All target scalar or SIMD-vector cores. This work extends the family along three axes: a bus-master DUT with role-inverted

agents, SIMT retirements (per-lane masked compare, split records, divergence), and weakly-ordered multi-core programs (the two-pass feed). Per-warp differential testing on this very DUT was concurrently demonstrated by FuzzGPU; our lockstep differs in the formalized alignment rules, the UVM packaging, and its role as one qualified checker inside a sign-off flow rather than a fuzzer’s oracle.

2.3. GPU kernel verification and testing

Software-level kernel verification is mature: GPUVerify [15] proves race- and divergence-freedom of OpenCL/CUDA kernels; GKLEE [16] performs concolic kernel testing, explicitly targeting divergent warps, non-coalesced accesses, and shared-memory bank conflicts—the same three phenomena our coverage layer quantifies at the RTL level; WEFT [17] verifies producer–consumer synchronization in GPU programs. Learned program generation has precedent in compiler fuzzing [18]. These operate on kernel source, not RTL; our contribution begins where their remit ends.

2.4. RTL GPU testing and memory consistency

FuzzGPU [3] (USENIX Security 2026) is the first RTL GPU fuzzer: SIMT-aware generation, trace-driven differential testing against ISA simulators on Verilator models of Vortex and Ventus, 20 bugs and 10 CVEs. Its evaluation is structural-coverage-only, its campaigns time-boxed, and it carries no functional-coverage model, no checker qualification, and no sign-off discipline—by design. Our methodology is the complementary half, and the flows cross-validate: the JALR LSB deviation we report (R1) was independently found by their S2 (filed against the project’s instruction-set simulator;

their RTL findings X1–X3, at a different pin, are disjoint from ours). Formal memory-consistency verification at RTL is addressed for CPU models by RTLCheck [21] and for GPU memory models in ASPLOS [22]; NVBitFI [20] brought systematic fault injection to GPUs for reliability studies. Concurrently, the Ventus project developed GVM, a UVM-like framework for its own design.

2.5. Checker qualification

Mutation-based qualification—proving checkers can fail—is established practice [19]; we apply the discipline end-to-end to a SIMT lockstep flow, which is what allows our pass rates and coverage to be read at face value.

3. The verification environment

[Ported text: role-inverted agents; DPI-C SimX; probes; end-state and lockstep checkers; assertion-aware verdicts; configurability; protocol assertions. Expanded treatment of the probe layer’s sampling semantics and of the launch-space sequences.]

4. Verdicts and non-vacuity

[Ported text.]

5. Per-instruction lockstep for SIMT: five rules, semi-formalized

5.1. The alignment problem

Let D be the DUT’s record stream per (core, warp), each record d carrying five fields (identifier, active-lane mask, PC, destination register, per-lane

values), and G the reference’s instruction stream g over the same fields, in program order. Comparing D and G naively produces false mismatches from five structural divergences. Each rule below is stated as *precondition* $\Rightarrow$ *soundness property*: if the precondition holds for the DUT, the transformation cannot mask a genuine defect (it may only remove false positives), and the residual comparator is exact outside the declared exception classes.

5.2. Rule 1 — retire order (sort by issue counter)

Precondition: the per-(core, warp) issue counter is strictly monotone in program order and does not wrap within a run. *Property:* sorting D by `uuid` per (core, warp) yields program order; hence record d_i aligns with instruction g_i by position. A retirement whose (PC, rd) differ from g_i remains visible. (Observed DUT behavior satisfies the precondition: an earlier-issued instruction was observed retiring after two later ones; the arbiter is `VX_commit.sv:56-71`.)

5.3. Rule 2 — cross-key (program-order keying)

Precondition: each stream independently enumerates the same per-(core, warp) instruction sequence. *Property:* aligning by per-(core, warp) ordinal position is well-defined without any shared identifier; the DUT `uuid`’s high bits supply (core, warp) attribution with no RTL change. Model-side identifiers (SimX leaves `uuid` at zero) are never used as keys.

5.4. Rule 3 — split retirements (mask-union aggregation)

Precondition: every record of one architectural instruction shares its `uuid`, and the union of the records’ thread masks equals the instruction’s

active-lane set L (records may overlap; observed `0xd` then `0xe`). *Property:* aggregating records by `uuid` with mask union, then comparing per active lane, is equivalent to comparing the instruction itself: every lane in L is compared exactly once semantically, and no lane outside L contributes.

5.5. Rule 4 — load data (own tap, region-filtered)

Precondition: (i) a tap where true post-extension load values are observable (the commit arbiter’s data field is stale for loads); (ii) a decidable predicate $V(a, v)$ that is true only when the two models’ memory states agree at address a (initialized region, no preceding divergent write, reference value not initialization poison). *Property:* comparing only lanes with V is a sound *under*-approximation: it never masks a defect (non- V loads are deferred to the byte-exact end-state compare) at the cost of comparing fewer sites. (Empirically: 429 false mismatches on a known-good kernel without the filter; 74 compared / 113 filtered / 0 false with it.)

5.6. Rule 5 — volatile CSRs

Precondition: a statically known CSR class whose values are timing-defined rather than architecturally defined (`mcycle/minstret/mhpmcounter*`). *Property:* excluding their *data* (still comparing PC and destination) removes only comparisons with no architectural ground truth; soundness is preserved, completeness is reduced by exactly that class.

5.7. Declared exception classes and the composite claim

Beyond Rules 4–5, the comparator declares one bounded, logged tolerance: the DUT’s `fsqrt.s` deviates by ≤ 1 ULP from correctly-rounded

IEEE 754 [14], tolerated only for `fsqrt` (same-sign, finite, ≤ 1 ULP via `fp32_within_ulp`), tallied in `n_fp_ulp_tol`, with a two-pass reconvergence feed re-checking all downstream operations bit-exactly. The composite soundness claim: *if the preconditions of Rules 1–3 hold and both models faithfully execute the same program, then any retirement whose (PC, destination, per-lane value) differs from the reference on an active lane, outside the declared exception classes, is reported as a mismatch*. The claim is conditional—the preconditions are DUT obligations an adopter must re-derive, listed as such in Section 12.

5.8. Validation

Lane-exact validation at five configurations (six programs) spanning a $2 \times 2 \times 3 \times 2$ axis space, including nested-divergence towers exercising Rule 3 through divergence and reconvergence; tallies and their provenance as in the conference version.

6. Verifying racy programs: the two-pass load-value feed

[Ported text with the assumption box and boundary discussion.]

7. Stimulus

[Ported text; simtgen axis status sentence updated after W4.]

8. A three-layer coverage model

[Ported text.]

9. Exclusion methodology: machine-generated, gate-checked waivers

[New text.]

10. Findings

10.1. RTL defects and hazards

[Ported text.]

10.2. Reference-model defects

[Ported text.]

11. Results

11.1. Closure campaign and bank provenance (W1)

The campaign concluded with a deliberate scope decision rather than a re-run for documentation-literalism: the divergence and memory closures are *in the suite bank* (`bank_1CL_1C_4W_4T_L2_20260909`: 23 covergroups, 504/524 bins = 96.18%, total 94.55% on the grown post-remediation coverage model); the barrier and VOTE/SHFL generator axes, built and verified by isolated merge (`simtgen_barrier_vote_shfl_20260910`), closed zero new bins because their targets (`cp_vote_shfl_op` 8/8, `cross_sfu_threads` 33/33) were already fully covered by dedicated directed kernels—an attribution correction to the initial reporting that we state rather than quietly keep. A relay-fixed 2CL bank (`bank_2CL_2C_4W_4T_relayfix_20260910`: 110 runs, 0 failures, 93.39% bins / 94.29% total) restores cross-configuration comparability on the fixed design. A third configuration point then closes the coverage story’s weakest waiver: shared L2/L3 had only ever run *passthrough-waived*

(the EUR exclusion class)—a waiver standing in for untested RTL. A seven-kernel cache-focused suite at 2CL with L2/L3 enabled on the relay-fixed design (all runs passing; the `cache_tier` phase is the project’s longest single run at $\sim 2\text{h}45\text{m}$) produced a dedicated bank whose liveness was verified recursively—real `cp_hit` counts on every new cache covergroup instance (L2: 86,503 and 88,104; L3: 36,501 and 81,854)—and, with the passthru waivers correctly not generated, a correspondingly enlarged denominator: 85.10% total (covergroup bins $1,087/1,680 = 64.70\%$). Expectations are set honestly: this is coverage and configurability evidence, not bug-finding—shared caches change timing, not architectural outcome, and the functional validation of L2/L3 predates it (57,379 matched lockstep pairs, 2026-08-07). With this bank, the coverage-run checklist that governed the campaign is fully closed, recorded in a final merged sign-off report. During this campaign the blocking hits-invariant waiver gate caught a real operator error—an exclusion set generated with 1CL keys against a 2CL merge—and aborted the merge; this is the third class of defect the gate has caught in operation, recorded here as evidence for the exclusion methodology of Section 9.

11.2. Verification cost (W2)

Lockstep’s wall-clock overhead is within run-to-run noise at every program size ($\pm 1.5\text{s}$ on $\sim 15\text{--}21\text{s}$ baselines; $\pm 20\text{s}$ —inside the inter-rep spread of plain mode—on the $\sim 860\text{s}$ kernel). Peak memory shows one real step: the second (SimX) model instance adds $\sim 29\%$ RSS at the large working set and $\sim 1.5\%$ at the small ones; the load-value feed adds no further measurable memory. The one FAIL in the grid is itself informative: `fpu\_test` under plain lockstep (feed disarmed) reproduces the known `fsqrt` 1-ULP de-

Table 1: Wall-clock and peak-RSS cost of the checking stack (median-of-two reps; QuestaSim 2021.2, 1CL/1C/4W/4T). Four programs spanning the kernel-size spectrum: `vecadd_lite` (~ 9.9 k cycles), `diverge_lite` (divergence kernel), `fpu_test` (FPU), `wide_stress` (256 KB high-toggle).

Program	Mode	Wall clock	Peak RSS
<code>vecadd_lite</code>	plain	15.3 s	296 MB
<code>vecadd_lite</code>	lockstep	15.0 s	299 MB
<code>vecadd_lite</code>	lockstep+feed	14.8 s	299 MB
<code>diverge_lite</code>	plain	17.8 s	297 MB
<code>diverge_lite</code>	lockstep	18.2 s	301 MB
<code>diverge_lite</code>	lockstep+feed	18.1 s	301 MB
<code>fpu_test</code>	plain	21.0 s	296 MB
<code>fpu_test</code>	lockstep	20.7 s	301 MB [†]
<code>fpu_test</code>	lockstep+feed	21.0 s	301 MB
<code>wide_stress</code>	plain	851 s	549 MB
<code>wide_stress</code>	lockstep	871 s	708 MB
<code>wide_stress</code>	lockstep+feed	856 s	708 MB

violation cascading into downstream mismatches—exactly the behavior Section 5’s exception-class design predicts—while the same program under lockstep+feed passes with residual zero: the reconvergence feed, not the tolerance alone, absorbs the cascade. A measurement artifact is disclosed rather than hidden: an initial `wide_stress` run used an undersized timeout budget, produced false TIMEOUT verdicts on all six rows, and was discarded (budget artifact, OBS-042 precedent); all reported rows completed with `rc=0`.

11.3. FuzzGPU PoC reproduction at our pin (W3-B)

FuzzGPU’s three Vortex RTL findings (filed at a different commit) were each exercised at our pin under this environment:

- **X1 (FPU compare-to-x0 unconditional writeback, PR #356): reproduced.** The PoC fires the RTL’s own `invalid writeback register` runtime assertion at the PoC’s exact `feq.s x0` PC; the defect is present and unfixed at our pin.
- **X2 (reserved M-extension funct7 decode, PR #358): reproduced—and re-scoped.** The upstream fix lives entirely in the reference model’s decoder; our RTL was always exact-match and never had the defect. A hand-encoded reserved opcode produces a genuine DUT-vs-SimX divergence in which the *hardware is right and the golden model is wrong* (DUT decodes ADD; SimX decodes MUL)—the paper’s central thesis (the reference model must be verified alongside the DUT) confirmed by an externally-sourced test case, and a caution against citing X2 as an RTL bug.
- **X3: static-only** (engineering cost of dynamic reproduction outweighed the value; documented, not attempted silently).

These runs are deliberately not suite members: X2’s expected outcome is a real mismatch, so it is kept as a directed qualification case instead.

11.4. Marginal coverage per generator kind (W3-A)

The ordering—directed > simtgen > riscv-dv—quantifies, on one axis, the claim the seed farm first drew for riscv-dv alone: the SIMT stimulus gap

needed a SIMT-aware generator, not more scalar seeds; and even a SIMT-aware generator cannot replace coverage-model validation (the bank-conflict and coalescing attribution lessons of Section 8).

11.5. *Bug-discovery dynamics (W7)*

Figure 1 plots cumulative register findings against project time, annotated with the methodology milestones that unlocked each cluster: the first lock-step bring-ups (reference-model defects became visible), the assertion-aware verdict gate (R10 surfaced the moment its silenced assertion was restored), the two-pass feed (racy-program findings), and the waiver gate (coverage-model defects). The curve is the paper’s process argument in one picture: checking depth, added in stages, kept producing findings on both sides of the comparison for the entire project—there was no point at which the methodology saturated into vacuous passes.

12. Threats to validity

12.1. *Internal validity*

The two-pass feed’s soundness rests on the assumption that a divergent in-region load is a genuine race resolution; we state it, discharge it with trace-level witnesses where possible, and classify rather than force-green elsewhere. Differential testing is structurally blind to stimulus faults (shared-binary blindness); one input class is closed by independent assertions, the class is reported open. Comparator tolerances are enumerated (load filter, volatile CSRs, fsqrt 1-ULP) and logged, never silent. Two infrastructure gaps are recorded rather than assumed away: no in-sim reset-testing methodology

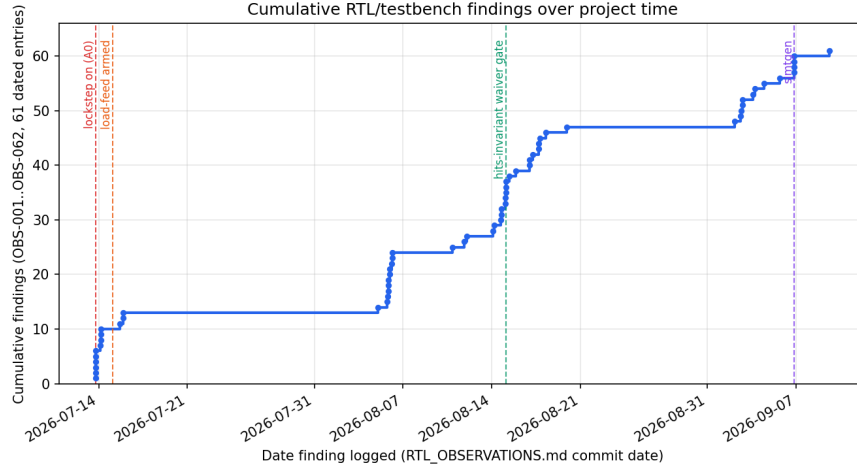


Figure 1: Cumulative findings (OBS register) over project time with methodology-milestone annotations. Data and reading: `figures/bug\_discovery\_curve.csv` / `.md` in the artifact.

exists (power-on reset only—the R10 bug class has never been exercised under randomized reset), and the environment has no isolated component-level test layer; both are open items, not silently covered claims.

12.2. External validity

All results are one DUT at one pin with locally modified files (disclosed), one simulator, one build; the alignment rules’ DUT preconditions are stated so adopters can re-derive them; transfer beyond is belief, not demonstration. Open-source UVM simulation is maturing (upstream UVM 2017 elaborates in Verilator with constrained randomization), but SystemVerilog functional coverage remains unsupported there, so the coverage-driven half requires a commercial simulator today.

12.3. Conclusion validity

Coverage figures are per-configuration and never blended; frozen banks are never overwritten; the D-extension elaboration is scoped by inspection *and* confirmed by an EXT_D-off recompile with a coverpoint-by-coverpoint bin diff (zero scored bins change; the entire delta is confined to two weight-0 coverpoints already excluded from every reported percentage); per-mnemonic claims avoid aliased encodings; bank provenance and design state accompany every table.

13. Toward silicon sign-off

[Ported text.]

14. Conclusion

[Ported and updated once Section 11 is filled.]

Acknowledgment

The authors thank the Vortex group at Georgia Tech and the maintainers of the open verification assets (core-v-verif, riscv-dv, RVVI, riscvISACOV, Spike) leveraged by the environment.

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Table 2: Marginal covergroup-bin contribution per program kind. Directed kernels are hand-targeted at an identified gap, so their per-program rate is not comparable on the same axis as the two generators; the two generator rows are measured directly.

Generator	Programs	Marginal contribution
riscv-dv (seed farm, seeds 2–11 $\times$ 9 profiles)	90	0 bins (370/377 unchanged; toggle +0.06% only)
simtgen (divergence + memory axes)	57	cp_split_depth 3/4 $\rightarrow$ 4/4; cp_bank_conflict 0/3 $\rightarrow$ 3/3 (credit shared with the OBS-060 probe reclassification)
simtgen (barrier + vote_shfl axes)	6	0 bins (targets already covered by directed kernels vote_shfl, bar_masks, sfu_masks)
Directed kernels (representative)	—	lmem_stress: scratchpad toggle 57.5% $\rightarrow$ 73.2%; multicore_isa: closed the per-core gap across all 4 cores; mshr_flood: 67,207 misses driven, target still unhit — itself finding OBS-031, not a closure